

Algebraic Geometry and Coding Theory

Lecture 1

September 10

Main theme. A code introduces redundancy so that valid transmitted words are separated in Hamming space. The central quantitative problem is to make a code simultaneously *large* and *well separated*. Reed–Solomon codes solve the finite-length problem optimally for their allowed lengths; asymptotic coding theory asks what remains possible when the alphabet is fixed and the block length tends to infinity.

Throughout, q denotes a prime power and $\mathbb{F}_q$ the finite field with q elements. We use row vectors for messages and codewords.

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1 Why introduce redundancy?

Suppose a message is transmitted through a channel and one or more symbols may be altered. If every possible word is regarded as a valid message, then after an error there is no internal evidence telling the receiver that anything went wrong. Coding theory deliberately restricts the set of valid transmitted words. The unused words create room in which errors can be detected or corrected.

Schematically,

$$\text{message} \longrightarrow \text{codeword} \longrightarrow \text{received word} \longrightarrow \text{decoded message}.$$

The encoder adds redundancy; the decoder uses the resulting structure.

1.1 ISBN-10: detection by one congruence

The historical ISBN-10 system is a useful first example. Write the ten symbols as

$$a_1, a_2, \dots, a_{10} \in \mathbb{F}_{11},$$

where the symbol X represents 10. The check condition can be written

$$\sum_{i=1}^{10} i a_i \equiv 0 \pmod{11}. \quad (1)$$

Since $10 \equiv -1 \pmod{11}$, the last symbol is determined by

$$a_{10} \equiv \sum_{i=1}^9 i a_i \pmod{11}.$$

For example, Walker's book has ISBN-10 0-8218-2628-X; the weighted sum of the first nine symbols is $208 \equiv 10 \pmod{11}$, so the check symbol is X .

Why does this detect a single-symbol substitution? If the symbol in position j changes by a nonzero amount $e \in \mathbb{F}_{11}$, then the checksum changes by je . Since $1 \leq j \leq 10$ and 11 is prime, both j and e are nonzero in $\mathbb{F}_{11}$, hence $je \neq 0$. Thus the altered word violates (1).

It also detects a transposition of two unequal symbols. Interchanging a_i and a_j changes the checksum by

$$i a_j + j a_i - i a_i - j a_j = (i - j)(a_j - a_i),$$

which is nonzero in $\mathbb{F}_{11}$ when $i \neq j$ and $a_i \neq a_j$.

Remark 1.1. The set of actual ISBN-10 strings is not literally a linear code over $\mathbb{F}_{11}$, because the first nine positions use decimal digits only. The ambient checksum space

$$\left\{ a \in \mathbb{F}_{11}^{10} : \sum_{i=1}^{10} i a_i = 0 \right\}$$

is a linear $[10, 9, 2]_{11}$ code, but this should be distinguished from the real-world identifier system.

The ISBN example is mainly a *detection* mechanism. One failed checksum does not generally tell us both where the error occurred and how it changed the symbol.

1.2 Repetition: correction at a high cost

The binary threefold repetition code is

$$C = \{000, 111\} \subset \mathbb{F}_2^3.$$

We encode 0 as 000 and 1 as 111. If at most one coordinate is corrupted, majority vote recovers the original bit.

For a k -bit message, repeating the whole block three times gives

$$u \mapsto (u, u, u) \in \mathbb{F}_2^{3k}.$$

This is a linear code of length $3k$, dimension k , and minimum distance 3. It corrects one arbitrary error, but its relative distance is $3/(3k) = 1/k$, which tends to zero as k grows. Repetition illustrates the basic tension: redundancy improves reliability, but too much redundancy destroys efficiency.

2 Codes, Hamming distance, and parameters

Definition 2.1. Let $\mathcal{A}$ be a finite alphabet. A *block code of length n* is a subset

$$C \subseteq \mathcal{A}^n.$$

Its elements are called *codewords*.

For coding over $\mathbb{F}_q$, the ambient space is $\mathbb{F}_q^n$.

Definition 2.2. For $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ in $\mathcal{A}^n$, the *Hamming distance* is

$$d_H(x, y) = \#\{i : x_i \neq y_i\}.$$

The Hamming weight of $x \in \mathbb{F}_q^n$ is

$$\text{wt}(x) = d_H(x, 0).$$

The *minimum distance* of a code with at least two codewords is

$$d(C) = \min_{x \neq y \in C} d_H(x, y).$$

Hamming distance is a metric. The triangle inequality follows coordinatewise: whenever $x_i \neq z_i$, at least one of $x_i \neq y_i$ or $y_i \neq z_i$ must hold.

2.1 Linear codes and generator matrices

Definition 2.3. A code $C \subseteq \mathbb{F}_q^n$ is *linear* if it is an $\mathbb{F}_q$ -vector subspace. If $\dim_{\mathbb{F}_q} C = k$ and $d(C) = d$, we say that C has parameters

$$[n, k, d]_q.$$

It contains exactly q^k codewords.

If $c_1, \dots, c_k$ is a basis of C , the matrix whose rows are $c_1, \dots, c_k$ is a *generator matrix* G . Encoding a message $u \in \mathbb{F}_q^k$ is simply

$$u \mapsto uG.$$

Lemma 2.4. *If C is linear, then*

$$d(C) = \min_{0 \neq c \in C} \text{wt}(c).$$

Proof. For $x, y \in C$, linearity gives $x - y \in C$, and

$$d_H(x, y) = \text{wt}(x - y).$$

Conversely every nonzero $c \in C$ occurs as $c - 0$. □

2.2 What minimum distance guarantees

For $x \in \mathbb{F}_q^n$ and $t \geq 0$, write

$$B_t(x) = \{y \in \mathbb{F}_q^n : d_H(x, y) \leq t\}$$

for the Hamming ball of radius t centered at x .

Proposition 2.5. *If a code has minimum distance d , then it can uniquely correct every pattern of at most*

$$t = \left\lfloor \frac{d-1}{2} \right\rfloor$$

errors.

Proof. If a received word y were within distance t of two distinct codewords c, c' , then

$$d_H(c, c') \leq d_H(c, y) + d_H(y, c') \leq 2t < d,$$

a contradiction. Thus the radius- t balls around distinct codewords are disjoint. $\square$

The proposition gives a worst-case guarantee. It does not say that every pattern with more than t errors must fail; only that unique correction is no longer guaranteed in general.

3 Parity checks and syndromes

Generator matrices describe a linear code by parametrizing its codewords. A second, dual description specifies the linear equations that codewords satisfy.

For a linear code $C \subseteq \mathbb{F}_q^n$, define

$$C^\perp = \{y \in \mathbb{F}_q^n : x \cdot y = 0 \text{ for every } x \in C\}.$$

If $\dim C = k$, then $\dim C^\perp = n - k$. A matrix H whose rows form a basis of $C^\perp$ is called a *parity-check matrix*. Then

$$C = \ker H = \{c \in \mathbb{F}_q^n : Hc^T = 0\}, \quad GH^T = 0.$$

If a generator matrix is in systematic form $G = [I_k \mid P]$, then one may take

$$H = [-P^T \mid I_{n-k}].$$

3.1 A small binary code

Consider

$$G = \begin{pmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \end{pmatrix}.$$

Encoding (u, v) gives

$$(u, v) \mapsto (u, v, v, u, u + v).$$

The four codewords are

$$00000, \quad 10011, \quad 01101, \quad 11110,$$

with weights 0, 3, 3, 4. Hence this is a $[5, 2, 3]_2$ code.

A parity-check matrix is

$$H = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{pmatrix}.$$

Definition 3.1. For a received word $y \in \mathbb{F}_q^n$, the *syndrome* is

$$s(y) = Hy^T.$$

If $y = c + e$ with $c \in C$, then

$$s(y) = H(c + e)^T = He^T.$$

Thus the syndrome depends only on the error vector modulo the code, not on the transmitted codeword.

For a one-coordinate error $e = ae_j$, its syndrome is a times the j th column of H . In the binary case, if the columns of H are nonzero and distinct, a one-bit error is located immediately by its syndrome.

For example, if 10011 is sent and 10111 is received, then

$$H(10111)^T = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix},$$

which is the third column of H . Under the promise of at most one error, the third coordinate is therefore the corrupted one.

4 The binary Hamming code

A particularly elegant use of syndromes is the binary Hamming code of length 7. Take

$$H_{\text{Ham}} = \begin{pmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{pmatrix}.$$

Its columns are precisely the seven nonzero vectors of $\mathbb{F}_2^3$. Since the last three columns form the identity matrix, H_{Ham} has rank 3, so

$$C = \ker H_{\text{Ham}}$$

has dimension $7 - 3 = 4$.

There is no weight-1 codeword because no column is zero. There is no weight-2 codeword because no two columns are equal. On the other hand, the first three columns sum to zero, giving a weight-3 codeword. Therefore

$$C \text{ has parameters } [7, 4, 3]_2.$$

A compatible systematic generator matrix is

$$G_{\text{Ham}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

Because the seven nonzero syndromes are exactly the seven columns of H_{Ham} , every single-bit error has a unique nonzero syndrome, and its location is read directly from that syndrome.

Moreover, the radius-1 balls around the $2^4 = 16$ codewords contain

$$16 \left(\binom{7}{0} + \binom{7}{1} \right) = 16 \cdot 8 = 128 = 2^7$$

words. Since the balls are disjoint by $d = 3$, they partition the whole ambient space. Such a code is called *perfect*.

5 Reed–Solomon codes

The Hamming code is built from a parity-check matrix. Reed–Solomon codes come from evaluating low-degree polynomials.

Lemma 5.1 (Root bound). *A nonzero polynomial of degree m over a field has at most m distinct roots, in that field or in any extension field.*

Proof. Induct on m . If a is a root, divide by $X - a$ to write $f = (X - a)g$ with $\deg g = m - 1$. Every other root of f is a root of g . $\square$

Fix distinct points $a_1, \dots, a_n \in \mathbb{F}_q$ and an integer $1 \leq k \leq n$. Let

$$V_k = \{f \in \mathbb{F}_q[X] : \deg f < k\} \cup \{0\}.$$

This is a k -dimensional vector space with basis $1, X, \dots, X^{k-1}$.

Definition 5.2. The Reed–Solomon evaluation code at $a_1, \dots, a_n$ is

$$\text{RS}_k(a_1, \dots, a_n) = \{(f(a_1), \dots, f(a_n)) : f \in V_k\}.$$

Evaluation is linear. If $f \in V_k$ evaluates to zero at all n points, then f has at least $n \geq k$ distinct roots while $\deg f < k$. By Lemma 5.1, $f = 0$. Thus the evaluation map is injective and the code has dimension k .

A generator matrix is the Vandermonde-type matrix

$$G_{\text{RS}} = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ a_1 & a_2 & \cdots & a_n \\ a_1^2 & a_2^2 & \cdots & a_n^2 \\ \vdots & \vdots & \cdots & \vdots \\ a_1^{k-1} & a_2^{k-1} & \cdots & a_n^{k-1} \end{pmatrix}.$$

Any nonzero $f \in V_k$ has at most $k - 1$ zeros among the evaluation points, so every nonzero codeword has weight at least

$$n - (k - 1) = n - k + 1.$$

Thus

$$d \geq n - k + 1.$$

We will shortly see that equality is forced by a general upper bound.

5.1 A concrete example over $\mathbb{F}_7$

Evaluate quadratics at $0, 1, 2, 3, 4 \in \mathbb{F}_7$. Then $n = 5$, $k = 3$, and

$$G_{\text{RS}} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 \\ 0 & 1 & 4 & 2 & 2 \end{pmatrix}.$$

The polynomial

$$f(X) = 1 + 2X + 3X^2$$

encodes as

$$(1, 6, 3, 6, 1).$$

A nonzero quadratic has at most two roots, so every nonzero codeword has weight at least 3. The polynomial $X(X-1)$ vanishes at exactly two of the five evaluation points, so the minimum distance is exactly 3. Hence this is a $[5, 3, 3]_7$ code.

Walker's original notation $\text{RS}(k, q)$ is the special case in which the evaluation points are all nonzero elements of $\mathbb{F}_q$, so $n = q - 1$.

6 The Singleton bound and MDS codes

Theorem 6.1 (Singleton bound). *Let $C \subseteq \mathcal{A}^n$ be a code over an alphabet of size q , with M codewords and minimum distance d . Then*

$$M \leq q^{n-d+1}.$$

In particular, a linear $[n, k, d]_q$ code satisfies

$$d \leq n - k + 1.$$

Proof. Delete any $d - 1$ coordinates from each codeword. The resulting map from C to $\mathcal{A}^{n-d+1}$ is injective: if two distinct codewords became equal, then they could differ only in the deleted $d - 1$ positions, contradicting minimum distance d . Therefore

$$M \leq |\mathcal{A}|^{n-d+1} = q^{n-d+1}.$$

For a linear code, $M = q^k$. □

A linear code meeting the Singleton bound with equality is called *maximum distance separable*, or *MDS*.

Corollary 6.2. *The Reed–Solomon code $\text{RS}_k(a_1, \dots, a_n)$ has parameters*

$$[n, k, n - k + 1]_q$$

and is MDS.

Proof. The root bound gave $d \geq n - k + 1$, while Singleton gives the reverse inequality. □

Thus Reed–Solomon codes are optimal for the parameters n and k : no code of the same length and size can have larger minimum distance.

7 The projective extension of Reed–Solomon codes

The affine line over $\mathbb{F}_q$ has q points. The projective line adds one point at infinity:

$$\mathbb{P}^1(\mathbb{F}_q) = \mathbb{F}_q \cup \{\infty\}, \quad \#\mathbb{P}^1(\mathbb{F}_q) = q + 1.$$

This gives one extra evaluation coordinate.

Let $1 \leq k \leq q + 1$ and write

$$f(X) = a_0 + a_1X + \dots + a_{k-1}X^{k-1}.$$

Define the extended evaluation vector by

$$((f(a))_{a \in \mathbb{F}_q}, a_{k-1}) \in \mathbb{F}_q^{q+1}. \tag{2}$$

The last coordinate is the “value at infinity” in the projective construction. Concretely, it is the top coefficient a_{k-1} .

If $\deg f = k - 1$, then the infinity coordinate is nonzero and f has at most $k - 1$ affine zeros. If $\deg f \leq k - 2$, then the infinity coordinate is zero but f has at most $k - 2$ affine zeros. Thus every nonzero codeword has at most $k - 1$ zero coordinates in total, so

$$d \geq (q + 1) - (k - 1) = q - k + 2.$$

The map is injective, and Singleton forces equality. Hence the projective extension is an

$$[q + 1, k, q - k + 2]_q$$

MDS code.

Remark 7.1. The extra coordinate in (2) is not obtained by naively substituting $X = \infty$ into a polynomial. In later algebraic-geometric language, one evaluates an appropriate section in a local trivialization at the point ∞ .

Reed–Solomon codes therefore have superb finite-length parameters, but over a *fixed* alphabet their length is bounded by $q + 1$. This is the first indication that long fixed-alphabet codes require a new idea.

8 General finite-length bounds

To compare arbitrary codes, including nonlinear ones, it is convenient to optimize directly over their size.

Definition 8.1. For $0 \leq d \leq n$, let

$$A_q(n, d) = \max\{|C| : C \subseteq \mathbb{F}_q^n \text{ has minimum distance at least } d\}.$$

The Singleton bound immediately gives

$$A_q(n, d) \leq q^{n-d+1}.$$

We now derive a different upper bound and an existence lower bound.

8.1 The Plotkin bound

Set

$$\theta = 1 - \frac{1}{q}.$$

Theorem 8.2 (Plotkin bound). *If $d > \theta n$, then*

$$A_q(n, d) \leq \frac{d}{d - \theta n}.$$

Proof. Let $C \subseteq \mathbb{F}_q^n$ have M codewords and minimum distance at least d . Consider the total Hamming distance over all ordered pairs of distinct codewords:

$$S = \sum_{x \neq y \in C} d_H(x, y).$$

Since every distinct pair has distance at least d ,

$$S \geq M(M-1)d.$$

For the opposite inequality, fix a coordinate j . For each $a \in \mathbb{F}_q$, let $m_{j,a}$ be the number of codewords having symbol a in that coordinate. The number of ordered pairs that differ in coordinate j is

$$\sum_{a \in \mathbb{F}_q} m_{j,a}(M - m_{j,a}) = M^2 - \sum_{a \in \mathbb{F}_q} m_{j,a}^2.$$

Since $\sum_a m_{j,a} = M$, Cauchy–Schwarz gives

$$\sum_a m_{j,a}^2 \geq \frac{M^2}{q}.$$

Hence the contribution of coordinate j is at most

$$M^2 \left(1 - \frac{1}{q}\right) = \theta M^2.$$

Summing over all n coordinates,

$$S \leq n\theta M^2.$$

Combining the two estimates,

$$dM(M-1) \leq n\theta M^2.$$

After dividing by M and rearranging,

$$(d - \theta n)M \leq d.$$

Since $d > \theta n$, division gives the result. $\square$

The hypothesis is essential: the formula is useful only when the relative distance exceeds the threshold $\theta = 1 - 1/q$.

8.2 Hamming balls and the Gilbert–Varshamov bound

The size of a Hamming ball depends only on its radius, not its center. Write

$$V_q(n, r) = |B_r(x)|.$$

To obtain a vector at distance exactly i from x , choose the i changed coordinates and then choose one of $q - 1$ new symbols in each. Therefore

$$V_q(n, r) = \sum_{i=0}^r \binom{n}{i} (q-1)^i. \quad (3)$$

Theorem 8.3 (Gilbert–Varshamov bound). *For $1 \leq d \leq n$,*

$$A_q(n, d) \geq \frac{q^n}{V_q(n, d-1)}.$$

Proof. Take a code C of maximum possible size $M = A_q(n, d)$. By maximality, every vector $y \in \mathbb{F}_q^n$ must lie in $B_{d-1}(x)$ for some $x \in C$. Otherwise y could be added to C while preserving minimum distance at least d .

Thus the radius- $(d-1)$ balls centered at the codewords cover the whole ambient space. They need not be disjoint, but

$$q^n \leq MV_q(n, d-1).$$

Rearranging proves the claim. $\square$

The nature of the two bounds is different. Plotkin says certain parameters are impossible; Gilbert–Varshamov says codes with certain parameters must exist.

9 Asymptotic coding theory

Finite-length optimality is not enough for the eventual goal of the course. We want to keep the alphabet $\mathbb{F}_q$ fixed and let the block length tend to infinity.

Suppose $C \subseteq \mathbb{F}_q^n$ has $M = q^k$ codewords. The quantity $k = \log_q M$ need not be an integer for a nonlinear code. Define

$$R = \frac{k}{n} = \frac{1}{n} \log_q M$$

to be the *information rate*, and

$$\delta = \frac{d}{n}$$

to be the *relative minimum distance*.

A useful family of long codes should keep both R and δ bounded away from zero.

9.1 The asymptotic rate function

Definition 9.1. For $0 \leq \delta \leq 1$, define

$$\alpha_q(\delta) = \limsup_{n \rightarrow \infty} \frac{1}{n} \log_q A_q(n, \lfloor \delta n \rfloor).$$

Informally, $\alpha_q(\delta)$ is the best asymptotic rate attainable while maintaining relative distance approximately δ .

The Singleton bound yields the immediate asymptotic estimate

$$\alpha_q(\delta) \leq 1 - \delta.$$

The Plotkin argument gives a stronger upper bound.

Theorem 9.2 (Asymptotic Plotkin bound). *Let $\theta = 1 - 1/q$. Then*

$$\alpha_q(\delta) \leq 1 - \frac{\delta}{\theta} \quad (0 \leq \delta \leq \theta),$$

and

$$\alpha_q(\delta) = 0 \quad (\theta < \delta \leq 1).$$

The proof of Theorem 9.2 was not part of this lecture and is omitted here.

9.2 The q -ary entropy function

Set $\theta = 1 - 1/q$ as above. For $0 \leq x \leq \theta$, define

$$H_q(x) = \begin{cases} 0, & x = 0, \\ x \log_q(q-1) - x \log_q x - (1-x) \log_q(1-x), & 0 < x \leq \theta. \end{cases}$$

This is the q -ary entropy function.

The exponential growth of Hamming balls is governed by H_q .

Lemma 9.3 (Asymptotic Hamming-ball volume). *For $0 \leq \lambda \leq \theta$,*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log_q V_q(n, \lfloor \lambda n \rfloor) = H_q(\lambda).$$

A proof follows from Stirling's formula; we use the lemma as an input.

Theorem 9.4 (Asymptotic Gilbert–Varshamov bound). *For $0 \leq \delta \leq \theta$,*

$$\alpha_q(\delta) \geq 1 - H_q(\delta).$$

Proof. For each n , monotonicity of $A_q(n, d)$ in d and Theorem 8.3 give

$$A_q(n, \lfloor \delta n \rfloor) \geq A_q(n, \lfloor \delta n \rfloor + 1) \geq \frac{q^n}{V_q(n, \lfloor \delta n \rfloor)}.$$

Taking $\log_q$, dividing by n , and passing to the limsup yields

$$\begin{aligned} \alpha_q(\delta) &\geq 1 - \lim_{n \rightarrow \infty} \frac{1}{n} \log_q V_q(n, \lfloor \delta n \rfloor) \\ &= 1 - H_q(\delta), \end{aligned}$$

by Lemma 9.3. □

Thus, for $0 \leq \delta \leq \theta$, the unknown optimal function is trapped between an explicit lower and upper bound:

$$1 - H_q(\delta) \leq \alpha_q(\delta) \leq 1 - \frac{\delta}{\theta}.$$

The lower inequality is constructive only in an existence sense: it tells us that sufficiently large codes with these asymptotic parameters exist. The upper inequality rules out rates above the Plotkin line.

10 Why algebraic geometry will enter

Reed–Solomon codes already show the basic mechanism that we will generalize:

- (1) choose a finite-dimensional space of functions;
- (2) evaluate those functions at many rational points;
- (3) control the number of zero coordinates by controlling how many zeros a nonzero function can have.

For polynomials of degree $< k$, the function space has dimension k , while a nonzero polynomial has at most $k - 1$ roots. This gives the MDS parameters $[n, k, n - k + 1]_q$.

The obstruction is length. Over a fixed field, the projective line has only $q + 1$ rational points, so Reed–Solomon evaluation cannot produce arbitrarily long codes over that same alphabet. The later geometric idea is to replace $\mathbb{P}^1$ by algebraic curves over $\mathbb{F}_q$ having many rational points, and to replace low-degree polynomials by Riemann–Roch spaces of rational functions. The asymptotic question then becomes a geometric one: can one find curves with many rational points compared with their genus?

This is the route from coding theory to algebraic geometry, and eventually to algebraic-geometric codes and the Tsfasman–Vlăduț–Zink improvement of the Gilbert–Varshamov lower bound.

Lecture summary

The main objects and conclusions from Lecture 1 are:

- A code is a subset of $\mathbb{F}_q^n$; its minimum Hamming distance controls guaranteed error correction.
- A linear $[n, k, d]_q$ code has q^k codewords and may be described either by a generator matrix G or a parity-check matrix H .
- Syndromes Hy^T record error information modulo the code; the $[7, 4, 3]_2$ Hamming code is a perfect one-error-correcting code.
- Reed–Solomon codes are polynomial evaluation codes and have parameters $[n, k, n - k + 1]_q$; they are MDS by the Singleton bound.
- Projective Reed–Solomon codes reach length $q + 1$ over $\mathbb{F}_q$.
- $A_q(n, d)$ measures the largest possible code size. Plotkin gives an upper bound, while Gilbert–Varshamov gives a lower bound.
- Asymptotically, $R = k/n$ and $\delta = d/n$ are measured by $\alpha_q(\delta)$. The Plotkin line is an upper bound and the Gilbert–Varshamov curve $1 - H_q(\delta)$ is a lower bound.

Reference for this lecture

- J. L. Walker, *Codes and Curves*, especially §1.1 and §§2.1–2.2.